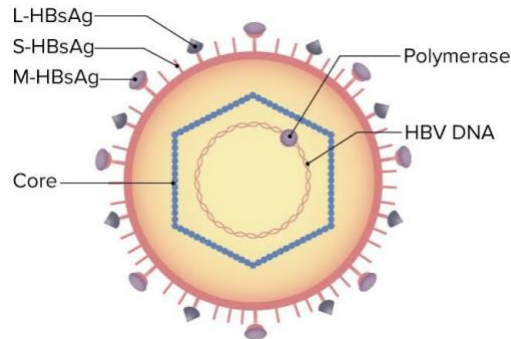


Q3. Describe clinical features, complications, diagnosis and management of serum hepatitis B or acute viral hepatitis B. Add a note on its prevention.



Viruses causing Acute Viral Hepatitis:
Specific hepatitis viruses:

- **Hepatitis A virus (HAV)**
- **Hepatitis B virus (HBV)**
- **Hepatitis C virus (HCV)**
- **Hepatitis D virus (HDV)**
- **Hepatitis E virus (HEV)**

Other viruses:

- **Cytomegalovirus**
- **Epstein-Barr virus**
- **Herpes simplex virus**

Hepatitis B

- **DNA virus**
- **Core** contains DNA and DNA polymerase enzyme for virus replication
- **Humans** are the only source of infection
- Transmitted by **infected blood, blood products, sexual contact, and from infected mother to child during delivery**
- **Tattooing and acupuncture** can also spread it

Route of Transmission

- **Horizontal transmission:**
 - Injection drug use
 - Infected unscreened blood
 - Tattooing, acupuncture needles
 - **Sexual** (homosexual and heterosexual)

- **Vertical transmission:**
 - **HbsAg-positive mother**

High-Risk Groups:

- **Haemodialysis patients, physicians, surgeons, dentists, paramedical staff, and laboratory/blood bank workers**
- Infected individuals are contagious during **incubation period** and as long as **HBsAg-positive**

Acute Hepatitis Clinical Features

1. Prodromal phase:

- **Anorexia, nausea, vomiting, headache, fatigue, malaise, myalgia, arthralgia**
- **Low-grade fever**

2. Icteric phase:

- **Jaundice** appears with dark urine and yellowish discoloration of skin and eyes
- **Clay-colored stool** and **pruritus** (suggests **cholestasis**)
- Some may never get jaundice (anicteric hepatitis)

3. Recovery phase:

- Improvement in symptoms and reduction in jaundice
- **Clinical and biochemical recovery** within 3-4 months

Signs of Acute Hepatitis:

- **Yellow sclera**
- **Skin scratch marks** due to **pruritus**
- **Tender hepatomegaly** in more than 50% cases
- **Splenomegaly** and **lymphadenopathy** in 10-20% cases

Complications of Acute Hepatitis:

- **Acute fulminant hepatitis**
- **Relapsing hepatitis**
- **Cholestatic hepatitis**
- **Chronic hepatitis** and **cirrhosis**
- **Aplastic anaemia**
- **Papular acrodermatitis**
- **Myelitis** and **neuropathy**

Investigations

Serology

- **HBV** has **3 important antigens**:
 - **HBsAg** (Hepatitis B Surface Antigen)
 - **HBcAg** (Core Antigen)
 - **HBeAg** (Hepatitis E Antigen)
- **HBsAg**:
 - Appears first in serum as evidence of infection
 - Persists for **3-4 weeks**, but can last up to **6 months**
 - After disappearance, **Anti-HBs** (antibody to HBsAg) appears and **confers immunity**
 - **Anti-HBs** means either **previous infection** or **vaccination**
- **HBcAg**:
 - Not found in blood
 - **Anti-HBc** appears early during illness
 - **IgM anti-HBc** indicates **acute infection**
 - **IgG anti-HBc** suggests **chronic infection** (when HBsAg is positive) or **recovery** (when Anti-HBs is positive)
- **HBeAg**:
 - Indicates **active viral replication** and high infectivity
 - **Anti-HBe** appears when HBeAg disappears, indicating **low viral replication** and decreased infectivity

Viral Blood

- **HBV-DNA** can be measured by **PCR** in the blood
- **Viral load > 105 copies/ml** indicates **active viral replication**

Other Investigations

- Significant increase in **ALT** and **AST** (>400 units/L) during early hepatitis
- **High alkaline phosphatase** suggests **cholestasis**
- **Prolongation of prothrombin time (PT)** indicates severe liver damage

Management

Acute Hepatitis

- **Supportive treatment** and monitoring for **acute liver failure**
 - **Physical activity** should be restricted, **bed rest** in severe cases
 - **High-calorie diet** with good **protein intake**

- **Hospitalization** and **IV fluids** (10% glucose) if oral intake is inadequate or severe nausea/vomiting
- Avoid **hepatotoxic drugs** or those metabolized in the liver
- **Cholestyramine** for **pruritus** in cases with **cholestasis**
- **Severe hepatic failure** (altered mental status, prolonged PT/bleeding) requires **hospitalization**
- No specific therapy for **acute viral hepatitis**, except for **acute HCV infection** (treat with **subcutaneous interferon alpha**)
- **Liver transplantation** may be required in **acute fulminant hepatic failure**

Prevention of Viral Hepatitis

- Prophylaxis available for **HAV** and **HBV**
- **Active immunization** (pre-exposure prophylaxis):
 - **Recombinant hepatitis B vaccine** containing **HBsAg**
 - **95% efficacy** in producing immunity in normal individuals

Q3. Describe the clinical features and management of Bronchial Asthma.



- Asthma is a **disease of the airways** characterized by **chronic inflammation** and **increased responsiveness** to various stimuli, causing **reversible airway obstruction**.
- Symptoms include **paroxysms of cough, wheeze**, and **dyspnoea**, with **symptom-free periods** in between.

Pathophysiology

- **Causes:** Combination of **genetic** and **environmental factors**.
- **Early onset (extrinsic):**
 - Starts in **childhood**, with a **family history of atopy** (e.g., allergic rhinitis, eczema).
 - **Positive skin test**, raised **IgE levels**.
- **Late onset (intrinsic):**

- Starts in **adulthood**, no family history of allergies.
 - **Negative skin test, normal IgE levels.**
- **Stimuli:** Allergic (e.g., dust, pollen) or non-allergic (e.g., cold air, exercise).
- **Bronchospasm:** Caused by allergen exposure leading to two-phase response:
 - **Early reaction (Type I): Histamine, leukotrienes** cause immediate bronchoconstriction.
 - **Late reaction (Type II): T-cell mediated inflammation**, involving **eosinophils**, leads to chronic bronchial inflammation.

Clinical Features

- **Episodic asthma:**
 - Sudden attacks of **dyspnoea, cough, and wheezing**; more common in children.
 - Episodes last **hours to days**; patients are asymptomatic between attacks.
- **Chronic asthma:**
 - Persistent symptoms in older adults, worse in the **early morning**.
 - May resemble **COPD**.
- Examination findings:
 - **Tachypnoea, tachycardia**, use of **accessory muscles**.
 - **Prolonged expiration**, audible **wheeze**.
- **Acute severe asthma** (life-threatening):
 - **Tachycardia, cyanosis, silent chest** on auscultation, **confusion or drowsiness**.

Investigations

- **Pulmonary function tests (PFT):**
 - **Hallmark:** Reversible airflow obstruction.
 - Reduced **FEV1, FEV1/VC ratio**, and **PEFR**. Improvement after bronchodilator confirms asthma.
- **Chest radiograph:** Usually normal; may show **hyperinflation**.
- **Peripheral blood:** Eosinophilia.
- **Sputum:** Increased **eosinophils**.

Management

General Measures

- **Avoid triggers:** Identify and avoid allergens or irritants.
- **Education:** Teach patients about asthma management, peak flow meter use, and inhaler techniques.

Drug Therapy

- **Quick relievers:**
 - **Inhaled short-acting beta agonists (e.g., salbutamol).**
- **Long-term control medications:**
 - **Inhaled corticosteroids:** Reduce inflammation.
 - **Mast cell stabilizers:** Prevent reactions to allergens.
 - **Long-acting beta agonists (e.g., salmeterol).**
 - **Leukotriene modifiers:** For additional control.
 - **Oral steroids:** For severe cases, lowest effective dose.

Management of Acute Severe Asthma

- **Hospitalization and oxygen therapy:** Maintain $\text{PaO}_2 > 60 \text{ mmHg}$.
- **Nebulized salbutamol/terbutaline:** Repeated every 20-30 minutes.
- **Ipratropium bromide:** Added for bronchodilation.
- **Systemic steroids:** **Prednisolone 30-60 mg/day.**
- **IV fluids:** To prevent dehydration.
- **Mechanical ventilation:** For respiratory failure.

Stepwise Therapy

- **Step 1:** Short-acting beta agonists as needed.
- **Step 2:** Add low-dose inhaled steroids.
- **Step 3:** Increase steroid dose, add long-acting beta agonist.
- **Step 4:** Add leukotriene modifiers or ipratropium.
- **Step 5:** Add regular oral steroids.

Complications

- **Pneumothorax.**
- **Respiratory failure.**
- **Cor pulmonale.**

Q1. Describe diagnosis, investigations and management of hypertension.

Or (essential hypertension and hypertensive retinopathy)

Or

Hypertensive patient in dental practice.

Hypertension is defined as a level of blood pressure where the risk for **target organ damage** increases, and the benefits of treatment outweigh the costs and hazards. Around **15%** of the general population can be considered hypertensive.

Classification of Hypertension

I. Primary (Essential) Hypertension

- Elevated blood pressure without any identifiable cause.
- **85-90%** of cases.
- **70%** of patients have a **positive family history**.
- **Multifactorial** pathogenesis: high salt intake, heavy alcohol use, obesity, sedentary lifestyle.

II. Secondary Hypertension

- Elevated blood pressure due to a specific, identifiable cause.
- Accounts for **15%** of cases.
- Causes:
 - **Renal disorders:** Renovascular stenosis, glomerulonephritis, polycystic kidney disease.
 - **Endocrinal disorders:** Pheochromocytoma, Cushing's syndrome, primary hyperaldosteronism, acromegaly, hypo/hyperthyroidism.
 - **Drugs:** Oral contraceptives, corticosteroids, sympathomimetics, NSAIDs, cyclosporine.
 - **Miscellaneous:** Toxaemia of pregnancy, coarctation of aorta, raised intracranial pressure, obstructive sleep apnoea.

Classification of Blood Pressure in Adults

Category	Systolic BP (mmHg)	Diastolic BP (mmHg)
Normal	<120	<80
Prehypertension	120-139	80-89
Stage 1 Hypertension	140-159	90-99
Stage 2 Hypertension	>160	>100

Clinical Features

- **Asymptomatic** in most cases, diagnosed on routine exams.
- **Symptoms due to high BP:** Occipital headache, dizziness, palpitation, fatigue.
- **Symptoms of target organ damage:** Epistaxis, hematuria, blurred vision, TIA, angina, breathlessness.
- **Symptoms of underlying causes:**
 - Weight gain (Cushing's syndrome).
 - Weight loss (thyrotoxicosis).

- Episodic headache, sweating (pheochromocytoma).

Physical Examination Findings

- **Truncal obesity** (Cushing's disease).
- **Palpable kidneys** (polycystic kidney disease).
- **Radiofemoral delay** (coarctation of aorta).
- **Recurrent back pain, undiagnosed fever** (chronic pyelonephritis).
- **Abdominal bruit** (renovascular disease).
- **Signs of complications:**
 - Heaving apex, 4th heart sound, loud aortic second heart sound.
 - Pulmonary crackles, retinal changes.

Risk Factors

- **Smoking, obesity, hyperlipidaemia, diabetes mellitus, family history of premature CAD.**

Complications of Hypertension

1. CNS Complications

- **TIA, CVA (stroke), subarachnoid hemorrhage.**
- **Hypertensive encephalopathy:** Very high BP with neurological symptoms (speech/vision disturbances, fits, loss of consciousness, papilloedema). Symptoms are usually reversible with BP control.

2. Ophthalmic Complications

- **Hypertensive retinopathy:** Thickening of retinal arterioles, narrowing of blood vessels, arteriovenous nipping, retinal hemorrhages, exudates, papilloedema.
 - **Grading of Retinopathy:**
 - **Grade I:** Arteriolar narrowing, increased light reflex.
 - **Grade II:** Marked arteriolar narrowing, arteriovenous nipping.
 - **Grade III:** Grade II + hemorrhages, exudates.
 - **Grade IV:** Grade III + papilloedema.

3. Cardiovascular Complications

- **Coronary artery disease** (angina, myocardial infarction).
- **Left ventricular failure.**
- **Aortic aneurysm, aortic dissection.**

4. Renal Complications

- **Proteinuria, progressive renal failure.**

5. Malignant Hypertension

- **Markedly high BP** with retinal hemorrhages/exudates, confusion, headache, vomiting, visual disturbances, and renal function deterioration.

Investigations in Hypertension

Basic Investigations for All Patients

- **Urine analysis:** Check for **protein, blood, and glucose.**
- **Blood urea nitrogen and serum creatinine:** Assess **renal function.**
- **Serum electrolytes:** Check **sodium and potassium** levels.
- **Blood glucose:** **Fasting and post-prandial** to assess **hyperglycaemia.**
- **Serum lipid profile.**
- **Serum calcium and uric acid.**
- **ECG:** Look for **left ventricular hypertrophy.**
- **Chest X-ray:** Check **cardiac size, cardiac failure, and aortic dilatation.**

Special Group of Patients

- **Chest X-ray and echocardiography.**
- **Renal ultrasonography and angiography.**
- **Serum calcium and phosphate.**
- **TSH** (Thyroid-stimulating hormone).
- **Urinary cortisol and catecholamines.**
- **Plasma renin activity and aldosterone.**

Treatment of Hypertension

1. General Measures (Lifestyle Modifications)

- **Relief of stress:** Avoid unnecessary tension; practice relaxation techniques.
- **Salt restriction:** Low sodium diet (**<100 mEq sodium** or **<6 g salt/day**).
- **Weight reduction:** Caloric restriction in obese patients to reduce BP.
- **Control of risk factors:** Restrict cholesterol and saturated fats, reduce oil intake, stop smoking, and moderate alcohol consumption.
- **Regular exercises, relaxation classes, meditation, and biofeedback.**

2. Antihypertensive Drug Therapy

Various classes of drugs are used to treat hypertension:

A. Diuretics

- **Thiazide diuretics:**
 - Chlorothalidone, **Chlorothiazide**, **Hydrochlorothiazide**, **Metolazone**.
- **Loop diuretics:**
 - **Furosemide**, **Bumetanide**.
- **Potassium-sparing diuretics:**
 - **Amiloride**, **Triamterene**.
- **Aldosterone receptor blockers:**
 - **Spirolonolactone**.

B. Beta-blockers

- **Propranolol**, **Metoprolol**, **Atenolol**, **Bisoprolol**.

C. Combined Alpha- and Beta-blockers

- **Carvedilol**, **Labetalol**.

D. ACE Inhibitors

- **Captopril**, **Enalapril**, **Lisinopril**, **Ramipril**, **Perindopril**.

E. Angiotensin II Antagonists

- **Losartan**, **Candesartan**.

F. Calcium Channel Blockers

- **Nifedipine**, **Verapamil**, **Diltiazem**, **Amlodipine**.

G. Alpha Blockers

- **Prazosin**, **Terazosin**, **Doxazosin**.

H. Direct Vasodilators

- **Hydralazine**, **Minoxidil**.

I. Central α_2 -blockers and Other Central Agents

- **Clonidine**, **Alpha-methyldopa**, **Reserpine**.

Diuretics:

- **Thiazides** are the most commonly used diuretics.

- **Side effects** of thiazides can be minimized by using a **low dose** (e.g., **25 mg/day** of hydrochlorothiazide).
- A combination of **thiazides** and **potassium-sparing agents** (e.g., **spironolactone, amiloride**) can be used.
- **Loop diuretics** are used less often.

Beta-blockers:

- Lower **heart rate** and **cardiac output**.
- **Cardioselective** beta-blockers (**atenolol, metoprolol**) are preferred.
- Used with caution in **heart failure, heart blocks, and asthma**.
- **Contraindications: COPD, asthma, heart failure, heart block, peripheral vascular disease, diabetes mellitus.**

ACE Inhibitors and Angiotensin Receptor Blockers (ARBs):

- **ACE inhibitors** (e.g., **enalapril, lisinopril**) and **ARBs** (e.g., **losartan, irbesartan**) have **good antihypertensive effects**.
- **ARBs** cause **fewer side effects** (no **cough** or **angioedema**) and are better tolerated.
- Contraindicated in patients with **renal impairment** and **bilateral renal artery stenosis**.

Calcium Channel Blockers:

- Act by causing **arterial vasodilatation**.
- Used in **angina, hypertensive crisis, Raynaud's disease, migraine, bronchial asthma, pulmonary hypertension, and congestive cardiac failure**.
- Common drugs: **Nifedipine, verapamil**.

Other Drug Groups:

- **Alpha adrenergic receptor blockers:** **Prazosin, doxazosin**.
- **Centrally acting agents:** **Clonidine, methyldopa**.
- **Vasodilators:** **Hydralazine, minoxidil**.
- **Mixed alpha- and beta-blockers:** **Labetalol, Carvedilol**.

Broad Guidelines for Drug Selection:

- **Diuretics** (thiazides) are often the first choice.
- **Beta-blockers** are useful in **young hypertensives** and those with **ischemic heart disease**.
- **Calcium channel blockers** are ideal for **elderly patients**, those with **asthma**, and **peripheral vascular disease**.

- **ACE inhibitors** are beneficial in patients with **diabetes, heart failure, and renovascular hypertension.**

Guidelines for Treatment Based on Hypertension Stage:

Stage of Hypertension	Drug Treatment
Prehypertension (without risk factors)	None
Prehypertension (with risk factors)	Diuretics
Stage 1 Hypertension	Beta-blockers, ACE inhibitors, Calcium channel blockers, Angiotensin receptor blockers, Thiazide diuretics
Stage 2 Hypertension	Thiazide diuretics + Beta-blockers, ACE inhibitors, Calcium channel blockers, Angiotensin receptor blockers. Add other drugs if BP is not controlled.

Q3. BASIC LIFE SUPPORT (BLS)

- **BLS:** Basic **manoeuvres** to prevent circulatory/respiratory arrest and support external circulation and ventilation without special equipment, except **AED** and protective shield.
- **Goal:** Provide **oxygen** and **perfusion** to brain and heart until advanced life support arrives.
- Most out-of-hospital **cardiac arrests** are due to **VF** or **pulseless VT, responsive to defibrillation.**
- BLS links for adult **chain of survival:**
 - Immediate **recognition** and activation of **ERS.**
 - Early **CPR.**
 - Rapid **defibrillation** (if needed).
- **Scene Safety:** Ensure **safety** before starting BLS.

IMMEDIATE RECOGNITION AND ACTIVATION OF ERS

- **Early recognition** of cardiac arrest is key; chances of survival decrease by **10% every minute.**

- If **unresponsive** and **not breathing normally** → assume **cardiac arrest**.
- Lone rescuer:
 1. Ensure **scene safety**.
 2. Check for **response** by tapping and shouting.
 3. If **unresponsive** with abnormal breathing → activate **ERS** and begin **CPR**.
- If **trained rescuer** → activate ERS and get **AED** while starting **CPR**.
- Lay rescuer:
 - No pulse check → assume **cardiac arrest** in unresponsive adults.
- **Healthcare provider: Check pulse for 10 seconds.**

EARLY CARDIOPULMONARY RESUSCITATION (CPR)

- **Steps of CPR: C-A-B** (Circulation, Airway, Breathing).

Circulation

- **Chest Compressions:**
 - **Forceful**, rhythmic compressions on **lower half** of sternum.
 - Generate **blood flow** to heart and brain.
 - Place **heel of one hand** in center of chest, other hand **overlap**.
 - Keep **arms straight**, shoulders over victim's chest.
 - **Rate: 100-120/minute.**
 - **Depth: 2-2.4 inches** (5-6 cm).
 - Allow **complete recoil** of chest.
 - **Ratio:** 30 compressions: 2 breaths (for all).
 - Avoid interruptions > **10 seconds** : Any pause in chest compressions should not last longer than 10 seconds
 - Change roles every **2 minutes** : Rotating the person performing chest compressions with another trained rescuer

Airway

- Open airway:
 - **Head tilt-chin lift** method for non-trauma victims.
 - For **suspected spinal injury** → **jaw thrust**.
 - If jaw thrust fails → use **head tilt-chin lift**.

Breathing

- **Mouth-to-mouth rescue breaths:**
 - Open airway, **pinch nose**, create airtight seal.
 - Give **1 breath over 1 second** → ensure **chest rise**.
 - Retry head tilt-chin lift if no rise.
 - **Ratio:** 30 compressions: 2 breaths.
- If **advanced airway** (e.g., endotracheal tube):
 - **1 breath every 6 seconds** (10 breaths/min).
 - No pause for compressions.

EARLY DEFIBRILLATION

- VF/pulseless VT → primary causes of cardiac arrest.
- Use **AED** immediately when available.
- Steps:
 - Place **electrode pads** on patient.
 - Stand **clear** during shock.
 - After shock → **start CPR** immediately for **5 cycles** (30:2) before rechecking rhythm.

ADVANCED LIFE SUPPORT (ALS)

- Includes **BLS** (C-A-B + defibrillation) with equipment, drugs, and advanced techniques.

Advanced Airway

- Use: **Endotracheal tube**, laryngeal mask, **cricothyrotomy**, etc.
- Confirm placement via **end-tidal CO2** monitoring.

Advanced Breathing

- Ventilation:
 - **Tidal volume:** 6-7 mL/kg.
 - **Rate:** **10 breaths/min.**
 - Avoid **excessive ventilation** (prevents complications).

Advanced Circulation

- Continue **chest compressions**.
- Use: IV access, cardiac monitoring, rhythm assessment, and drugs.
- With **advanced airway** → no compression-pause; **rate**: 100-120/min, ventilation: **1 breath/6 seconds**.

SHOCKABLE RHYTHMS: VF/PULSELESS VT

- Defibrillation:
 - **Monophasic**: 360 J
 - **Biphasic**: 150-200 J
- After shock → **start CPR** immediately.
- **Adrenaline**: 1 mg IV every **3-5 minutes**.
- **Amiodarone**:
 - **First dose**: 300 mg IV.
 - Repeat: 150 mg if persists.
- If no amiodarone → **Lidocaine** (max **3 mg/kg**).

NON-SHOCKABLE RHYTHMS: PEA/ASYSTOLE

- Continue **CPR** for 2 minutes.
- Give **adrenaline**: 1 mg IV every **3-5 minutes**.
- Recheck **rhythm** after every **2 minutes**.
- If rhythm changes → check **pulse**:
 - **Pulse present** → start post-resuscitation care.
 - **Pulse absent** → continue CPR.

METRONIDAZOLE

Dosage:

- **800 mg t.d.s.** for intestinal infections.
- **400 mg t.d.s.** for extra-intestinal infections.
- **Duration**: 10 days.

Mechanism of Action:

- **Metronidazole** is a **nitroimidazole** antibiotic that is primarily effective against **anaerobic** bacteria and **protozoa**.
- The drug works by entering the **microorganism** and getting reduced by the **electron transport proteins** inside the cell.

Q2. Define acute nephritic syndrome. Discuss the clinical features and treatment of poststreptococcal acute glomerulonephritis

Or

What are the causes and management of glomerulonephritis

Or

Describe the etiology, clinical features, complications, investigations and management of acute nephritis.

- A clinical syndrome caused by **acute glomerular inflammation**.
- Characterized by **oliguria, edema, hypertension, hematuria, subnephrotic proteinuria** (<3 g/day), and **worsening renal function**.

Aetiology poststreptococcal acute glomerulonephritis/ acute nephritis.

- **Post-streptococcal (most common):** Group A β -hemolytic streptococcus.
- **Other causes:**
 - **Henoch-Schonlein purpura**
 - **Hemolytic uremic syndrome**
 - **Severe IV hemolysis**
 - **Infections:** Hepatitis B, subacute bacterial endocarditis (SABE)
 - **IgA nephropathy**
 - **Familial nephropathy**
 - **Non-streptococcal causes:** Staphylococcal, salmonella, or pneumococcal infections

Clinical Features

In Children:

- **Abrupt onset**
- **Hematuria:** Frank blood in urine, oliguria
- **Facial puffiness**
- **Hypertension**
- **Pulmonary edema** due to **left ventricular failure**
- **Nausea, vomiting** (due to raised blood urea)
- **Preceding infection history**

In Adults:

- **Slow onset**
- **No preceding infection history**

- **Smoky urine** (oliguria)
- **Hypertension**
- **Pulmonary edema** absent
- **Nausea, vomiting**

Signs

- **Raised BP**
- **Pitting pedal edema**
- **Respiratory:** Bilateral fine crepitations
- **Facial puffiness**

Investigations

- **Urine Analysis:**
 - Non-selective proteinuria
 - Globulin and albumin casts
 - RBC and granular casts under the microscope
- **Blood tests:**
 - Elevated **blood urea** and **serum creatinine**
 - Total proteinuria <3 g/24 hours
 - **Potassium** may be raised in renal failure

Treatment

- **Bed Rest:** Until symptoms improve.
- **Dietary Restrictions:**
 - Low **protein, sodium, and potassium**.
 - Restrict **fluid intake:** Calculate daily as urine output (ml) + 500 ml.
 - Avoid juices.
- **Hypertension Management:**
 - **Hydralazine, beta-blockers** (e.g., atenolol), or **calcium channel blockers**.
- **Antibiotics:**
 - **Penicillin:** Procaine penicillin IM for 6 days to eradicate streptococci.
- **Diuretics:**
 - For acute **LVF** or **pulmonary edema**. Drug of choice: **Furosemide**.
- **Severe Cases:** Dialysis may be required in progressive renal failure.

Complications:

- Congestive heart failure
- Pulmonary oedema
- Chronic glomerulonephritis
- Nephrotic syndrome

- This reduction process activates the **nitro group** of metronidazole, making it toxic to the microorganism. The activated compound interacts with the **DNA** of the pathogen, causing **DNA strand breakage** and inhibiting **DNA synthesis**.
- As a result, the microorganism is unable to **replicate** and **repair** its DNA, leading to **cell death**.
- The drug is particularly effective against **anaerobic bacteria** and **protozoa**, both of which are reliant on anaerobic metabolism.

Indications:

- **Giardiasis:** Infection caused by the **protozoan Giardia lamblia**. Metronidazole is used to treat gastrointestinal symptoms caused by the parasite.
- **Trichomonas vaginitis:** A **sexually transmitted infection** caused by the protozoan **Trichomonas vaginalis**. Metronidazole helps in clearing the infection.
- **Dracunculosis:** Caused by the **Guinea worm** (*Dracunculus medinensis*), where metronidazole is used in some cases to treat complications.
- **Ulcerative gingivitis:** A **gum infection** caused by anaerobic bacteria. Metronidazole is useful due to its **antibacterial** action.
- **Amoebiasis:** **Amoebic infections** caused by **Entamoeba histolytica** that affect the intestines and extra-intestinal sites, such as the liver.
- **Anaerobic infections:** Effective in treating **anaerobic bacterial infections** that occur in wounds, abscesses, or surgical infections.

Side Effects:

1. **Gastrointestinal (GI):**
 - **Nausea, vomiting, diarrhea.**
 - **Metallic taste** in the mouth.
 - **Abdominal pain** or discomfort.
2. **Central Nervous System (CNS):**
 - **Headache, dizziness, and vertigo.**
 - **Ataxia** (lack of muscle coordination).
3. **Skin and Allergic Reactions:**
 - **Urticaria** (hives), **pruritus** (itching).
 - **Flushing** (redness of the skin).
4. **Antabuse-like Reaction:**
 - If consumed with alcohol, metronidazole causes an **Antabuse-like** reaction, leading to symptoms like **nausea, vomiting, flushing, and headache**. This is due to the drug's ability to inhibit **acetaldehyde dehydrogenase**, leading to **accumulation of acetaldehyde** when alcohol is consumed.

Important Notes:

- Metronidazole should be taken with caution in patients with **liver disease** due to its metabolism in the liver.
- It is contraindicated during **pregnancy** (especially in the **first trimester**) unless absolutely necessary.
- **Alcohol** should be avoided during treatment due to the **Antabuse-like** reaction.

In summary, metronidazole is a broad-spectrum **antiprotozoal** and **antibacterial** agent that is primarily used for treating infections caused by **anaerobic organisms** and certain **protozoa**. Its **mechanism of action** involves the reduction of its nitro group, leading to damage of the **DNA** of the pathogen, which prevents replication and results in **cell death**. It is widely used for a variety of infections, but patients should be aware of its potential side effects and the need to avoid alcohol during treatment.

Q10. Hypothyroidism

- **Deficiency** of thyroid hormones.
- **Primary:** Thyroid gland disorder.
- **Secondary:** Decreased TSH from pituitary.
- **Myxoedema:** Severe hypothyroidism with mucopolysaccharide accumulation, causing swollen skin, thickened facial features.

Hyperthyroidism

- **Excess T4/T3** hormones in circulation.

Causes of Hypothyroidism

- **Primary Hypothyroidism:**
 - Autoimmune (spontaneous atrophic).
 - **Iodine deficiency.**
 - **Post-131I therapy, thyroidectomy.**
 - **Hashimoto's thyroiditis.**
 - Genetic defects (dyshormonogenesis).
 - Drugs (lithium, amiodarone, etc.).
 - Radiation.
- **Secondary Hypothyroidism:**
 - Pituitary gland destruction (surgery, tumors).
 - Hypothalamic disorders.

Clinical Features

1. **General:** Weakness, cold intolerance, weight gain, puffy face, dry skin, hoarseness, goiter.
2. **GI:** Low appetite, constipation, ascites.
3. **Cardio:** Bradycardia, hypertension, cardiac failure.
4. **Neuromuscular:** Muscle stiffness, delayed reflexes, depression.
5. **Skin:** Dry flaky skin, alopecia, vitiligo, myxoedema.
6. **Reproductive:** Menorrhagia, infertility, impotence.
7. **Hematology:** Anemia, macrocytosis.
8. **Others:** Tiredness, slurred speech, cold intolerance.

Investigations

1. **TSH:** High in primary, low in secondary.
2. **T3/T4:** Low levels.
3. **Antibodies:** High TPO, thyroglobulin in Hashimoto's.
4. **Other labs:** High cholesterol, triglycerides, creatine kinase, hyponatremia.

Treatment

- **Levothyroxine:** Start with 50-100 µg/day.
 - Adjust to maintain **TSH <3 mU/L**.
 - Elderly/heart disease: Start with 25 µg/day.
- **Lifelong therapy.**

Dental Considerations

- Risk of **bleeding, slow healing**.
- Avoid sedatives, opioids (risk of myxoedema coma).
- Controlled hypothyroidism: Safe for dental procedures.

Q13. Thrombocytopenia

Definition:

Thrombocytopenia refers to a **decrease in the number of platelets** in the blood, with a count below **150,000/µL**, leading to an increased risk of bleeding.

Thrombocytopenia Causes

1. Decreased Platelet Production

- **Bone Marrow Disorders:** Conditions like **leukemia**, **aplastic anemia**, and **myelodysplastic syndromes** can damage the bone marrow, leading to fewer platelets being produced.
- **Nutritional Deficiencies:** Lack of **vitamin B12 or folic acid** can impair platelet production.
- **Viral Infections:** Viruses like **HIV**, **hepatitis**, and **parvovirus** can affect the bone marrow and reduce platelet count.
- **Chemotherapy and Radiation:** These treatments can suppress bone marrow, resulting in thrombocytopenia.

2. Increased Platelet Destruction

- **Autoimmune Disorders:** Conditions like **immune thrombocytopenic purpura (ITP)** cause the immune system to attack platelets.
- **Drugs:** Certain medications like **heparin** or **quinine** can trigger immune reactions that destroy platelets.
- **Infections:** Severe infections like **sepsis** can lead to widespread destruction of platelets.
- **Disseminated Intravascular Coagulation (DIC):** This condition involves excessive clotting throughout the body, leading to platelet consumption and thrombocytopenia.

3. Platelet Sequestration

- **Splenomegaly:** Enlargement of the spleen, often due to conditions like **cirrhosis**, **lymphoma**, or **malaria**, can cause a large number of platelets to become trapped in the spleen, leading to low circulating platelet counts.

4. Dilutional Thrombocytopenia

- **Massive Blood Transfusion:** When a large amount of blood is transfused, it may dilute the platelets, resulting in low platelet counts.

5. Genetic Disorders

- **Wiskott-Aldrich Syndrome:** A rare inherited disorder causing thrombocytopenia, eczema, and immune deficiency.
- **Bernard-Soulier Syndrome:** A rare genetic condition where platelets are larger than normal and less effective, leading to low platelet counts.

Clinical Features:

1. **Skin manifestations:** Petechiae, purpura, ecchymoses
2. **Mucosal bleeding:** Nosebleeds (epistaxis), gum bleeding
3. **Excessive bleeding after trauma** or surgery
4. **Menorrhagia:** Heavy menstrual bleeding
5. **Hematuria:** Blood in urine

Investigations:

1. **Complete Blood Count (CBC):** Platelet count below normal
2. **Peripheral Blood Smear:** To assess platelet morphology
3. **Bone Marrow Examination:** For decreased production causes
4. **Coagulation Profile:** PT, aPTT, fibrinogen levels
5. **Specific Tests:** Antiplatelet antibodies, viral markers for HIV or hepatitis

Management:

1. **Treat the underlying cause**
2. **Platelet transfusions:** For severe thrombocytopenia ($<10,000/\mu\text{L}$)
3. **Immunosuppressive therapy:** Steroids in immune causes like ITP
4. **Splenectomy:** In refractory cases
5. **Medications:** Thrombopoietin receptor agonists (eltrombopag, romiplostim)
6. **Avoidance of anticoagulants** or NSAIDs

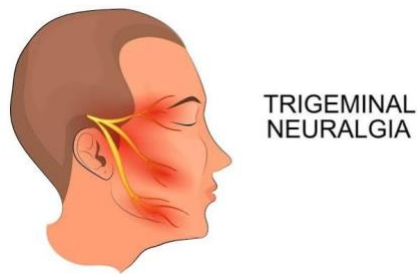
Complications:

1. **Severe bleeding:** Intracranial hemorrhage
2. **Infections:** In bone marrow failure syndromes
3. **Iron deficiency:** Due to chronic bleeding

Prognosis:

Depends on the underlying cause and response to treatment. Most cases of ITP resolve with therapy, while thrombocytopenia in DIC requires aggressive management.

Q8. Mention different types of neuralgias. Explain in detail about trigeminal neuralgia.



Neuralgias:

Various types of facial and cervical neuralgias are as follows:

- **Trigeminal neuralgia**
- **Raeder's paratrigeminal neuralgia**
- **Atypical pain/neuralgia**
- **Geniculate neuralgia**
- **Glossopharyngeal neuralgia**
- **Migrainous neuralgia**
- **Occipital neuralgia**
- **Post-herpetic facial neuralgia**
- **Sphenopalatine ganglion neuralgia**
- **Superior laryngeal neuralgia**
- **Tympanic plexus neuralgia**

Trigeminal Neuralgia:

- **Trigeminal neuralgia** is a disorder of the **trigeminal nerve** (5th cranial nerve). It is also known as **tic-douloureux**, **Fothergill's disease**, or **trifacial neuralgia**.

Aetiology (Causes):

- Most cases are **idiopathic** (unknown cause).
- It may occur due to **pressure** on the trigeminal nerve from conditions like **tumours** or **vascular anomalies**.
- **Circulatory insufficiency** or **reflex vasoconstriction** of the **Gasserian ganglion**.
- An area of **demyelination**, as seen in patients with **multiple sclerosis**, may also trigger it.

Clinical Features:

- More common in **older adults** than in younger individuals.
- Pain is confined to the **area of distribution** of the trigeminal nerve.

- **Right side** of the face is more commonly affected than the left side.
- The pain is **searing, stabbing, or lancinating**, and lasts only a few seconds or minutes. It is **unilateral** and rarely crosses the midline.
- **Tic-douloureux** applies when the patient experiences spasmodic contractions of the **facial muscles**.
- A '**trigger zone**' is characteristic of trigeminal neuralgia. These zones, when touched, precipitate an attack of pain. Common trigger zones include the **vermillion border of the lips, ala of the nose**, and around the eyes.
- **Triggering activities**: Touching, laughing, eating, exposure to a strong breeze, or simply smiling can trigger pain.

Differential Diagnosis:

- **Migraine or migrainous neuralgia** (Horton's syndrome, histamine headache).
- **Post-herpetic neuralgia**.
- **Costen's syndrome**.
- **Trotter's syndrome**.

Treatment:

1. **Medical Management:**
 - **Carbamazepine** (up to **600 to 1200 mg/day**): Commonly used as a diagnostic and therapeutic challenge.
 - **Phenytoin (Dilantin)** (100 mg TDS): Effective in some cases.
 - **Baclofen** (50 to 60 mg/day): An anticonvulsant often used.
2. **Injection Therapy:**
 - Injection of **alcohol** or **boiling water** into the peripheral nerve area or the **Gasserian ganglion** can provide pain relief.
3. **Surgical Treatment:**
 - **Peripheral neurectomy**: Involves sectioning the nerve at the **mental foramen, supraorbital, or infraorbital foramen**.
 - **Microsurgical decompression** of the trigeminal nerve root is a newer procedure with good results for managing trigeminal neuralgia.

Q2. Describe aetiopathogenesis, clinical features and treatment of peptic ulcer.



- **Definition:** Ulcer in lower oesophagus, stomach, duodenum, jejunum (post-surgical anastomosis), or ileum near Meckel's diverticulum
- **Cause:** Imbalance between mucosal defense and luminal factors (acid and pepsin)
- **Characteristics:** Breach >5 mm deep into submucosa
- **Common Types:**
 - a. **Duodenal ulcer:** Common, affects younger age (30-55 years)
 - b. **Gastric ulcer:** Less common, affects older age (55-70 years)

Aetiology peptic ulcer

- **Hereditary:** Strong family history, especially in gastric ulcers
- **Helicobacter pylori:**
 - a. Main cause (70% gastric ulcers)
 - b. Produces mucosal damage via urease, catalase, lipase, adhesins
 - c. Converts urea to ammonia, causing mucosal damage
- **NSAIDs:**
 - a. Decrease prostaglandin levels, impair cytoprotection
 - b. Cause mucosal injury, erosions, ulcers
- **Smoking:** Delays healing, increases complications
- **Corticosteroids:** Silent perforation
- **Acid-Pepsin vs. Mucosal Resistance:**
 - a. Gastric hypersecretion (common in Zollinger-Ellison syndrome)
 - b. Acid important in duodenal ulcers
- **Factors Reducing Mucosal Resistance:**
 - a. Drugs (aspirin, NSAIDs)
 - b. H. pylori
 - c. Bile reflux from weak pyloric sphincter
- **Other Risks:** Smoking, alcohol

Clinical Features

- **Chronic Condition:** Relapses and remissions over years
- **Abdominal Pain:**
 - a. Localized to epigastrium (pointing sign)
 - b. Burning, hunger pain (relieved by food/antacids)
 - c. Night pain (wakes at 3 AM, relieved by food)
 - d. Episodic pain (lasts 1-3 weeks, occurs 3-4 times a year)
- **Seasonal Relapses:** More in winter and spring
- **Other Symptoms:**
 - a. Water-brash, heartburn, vomiting, nausea, dyspepsia
 - b. Haematemesis (more common), weight loss, anaemia

Complications

- Haematemesis
- Gastric outlet obstruction
- Acute perforation
- Chronic blood loss anaemia

Investigations

- **Barium Meal:** Shows ulcer as a crater or deformed cap
- **Endoscopy:** Visualizes ulcer, biopsy rules out malignancy and *H. pylori*
- **Serum Gastrin and Gastric Acid:** For Zollinger-Ellison syndrome
- ***H. pylori* Tests:**
 - a. **Invasive:** Rapid urease test, histology, culture from biopsy
 - b. **Non-invasive:** Serology, urea breath test

Treatment of Peptic Ulcer

I. Short-term Management

Drugs used:

1. **Antacids & Alginates**
 - Examples: Aluminium hydroxide, magnesium trisilicate, sodium bicarbonate.
 - **Action:** Forms a protective mucosal "raft".
 - **Dose:** 15-30 ml liquid after meals & at bedtime for 4-6 weeks.
 - **Use:** Symptomatic relief.
2. **H₂-receptor Antagonists**
 - Examples: Cimetidine, ranitidine, famotidine.
 - **Action:** Inhibit H₂ receptors on parietal cells.
 - **Dose:**
 - Cimetidine: 400 mg BD or 800 mg at night.
 - Ranitidine: 150 mg BD or 300 mg at night.
 - Famotidine: 20 mg BD or 40 mg at night.
 - **Duration:**
 - Duodenal ulcer: 4 weeks.
 - Gastric ulcer: 6 weeks + follow-up endoscopy.
3. **Proton Pump Inhibitors (PPIs)**
 - Examples: Omeprazole, lansoprazole, pantoprazole, esomeprazole, rabeprazole.
 - **Action:** Irreversibly inhibits H⁺/K⁺ ATPase (proton pump).
 - **Dosage:**
 - Omeprazole, rabeprazole: 20 mg daily.
 - Lansoprazole: 15-30 mg daily.

- Esomeprazole: 20-40 mg daily.
 - Pantoprazole: 40 mg daily.
 - **Indications:** GERD, Zollinger-Ellison syndrome, H. pylori adjunct therapy.
 - **Note:** Take 30 mins before meals (except rabeprazole/pantoprazole).
- 4. **Prostaglandin Analogues**
 - Example: Misoprostol (200 mcg QID).
 - **Action:** Enhances mucosal protection.
 - **Use:** Prevent NSAID-induced ulcers.
- 5. **Colloidal Bismuth Compounds**
 - Examples: Bismuth subsalicylate, bismuth subcitrate.
 - **Action:**
 - Forms protective coat over ulcer.
 - Antimicrobial against H. pylori.
- 6. **Complex Salts**
 - Example: Sucralfate.
 - **Action:** Coats ulcers, protects from acid-pepsin digestion.

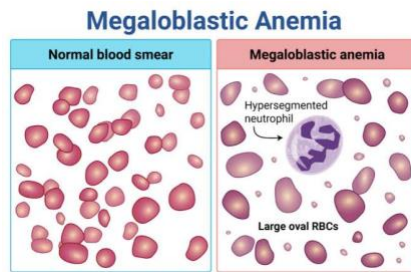
H. Pylori Eradication Therapy

- **Triple Therapy:**
 - PPI + Amoxicillin (1 g BD) + Clarithromycin (500 mg BD) for 14 days.
 - Metronidazole replaces amoxicillin if allergic.
- **Quadruple Therapy** (if triple therapy fails):
 - PPI + Bismuth + Tetracycline + Metronidazole.

II. Long-term Management

1. **Intermittent Treatment:**
 - For relapses <4 times/year.
 - Use ulcer healing agents for 4 weeks.
2. **Maintenance Treatment:**
 - Rarely needed post-H. pylori eradication.
 - Lowest effective dose of H₂ blockers if required:
 - Cimetidine: 400 mg at night.
 - Ranitidine: 150 mg at night.
3. **Surgical Treatment:**
 - **Gastric ulcer:** Partial gastrectomy (Billroth I anastomosis).
 - **Duodenal ulcer:** Truncal vagotomy + pyloroplasty or gastroenterostomy.
 - **Emergency surgery:** For bleeding or perforation.
 - **Elective surgery:** For gastric outlet obstruction or recurrent ulcers.

Q2. Describe the aetiological factors, clinical features and management of Megaloblastic Anaemia. Describe the differential diagnosis of Megaloblastic anaemia.



Megaloblastic anaemia results from **deficiency of vitamin B12 or folic acid** or disturbances in folic acid metabolism.

- **Megaloblasts** are RBCs that are **abnormally large** and nucleated but well haemoglobinated.
- **Macrocytes** are erythrocytes with increased **mean corpuscular volume (MCV)**, derived from megaloblasts.
- Thus, **megaloblastic macrocytic anaemia** refers to the features in both bone marrow and peripheral blood.

Aetiology

- Most megaloblastic macrocytic anaemias are caused by **vitamin B12** or **folate deficiency** or both.
- Both vitamin B12 and folate are essential for **DNA synthesis**.
- **Deficiency** of either or both results in **failure of DNA synthesis**, causing abnormal cell proliferation, especially in haemopoietic tissues and gastrointestinal epithelial cells.
- Impaired DNA synthesis leads to delayed **cell division**, larger cell size (megaloblasts), but normal **haemoglobin synthesis**.
- These large erythroid cells are often destroyed in the bone marrow, releasing **lactate dehydrogenase (LDH)** into the blood.
- In bone marrow, **giant metamyelocytes** and **dysplastic megakaryocytes** are also seen.

Clinical Features

- Clinical manifestations affect the **haematological, gastrointestinal, and nervous systems**.
- **Anaemia** with possible **mild jaundice** due to raised **unconjugated bilirubin**.
- Gastrointestinal symptoms: **splenomegaly, anorexia, weight loss, diarrhoea, smooth and beefy red tongue**.

- Neurological manifestations in **cobalamin deficiency**: **paraesthesia, ataxia, sensory-motor paraparesis, forgetfulness, psychosis.**

Investigations

- **RBC count, reticulocyte count, total leucocyte count, and haemoglobin** are reduced.
- **Platelet count** is reduced (**thrombocytopenia**).
- **Indirect bilirubin** mildly elevated.
- **Serum iron** is elevated; **iron binding capacity** is reduced.
- **Serum ferritin** level is increased.
- **Plasma LDH** is markedly increased.
- Specific tests for **B12/folate deficiency**.
- **Peripheral blood smear: macrocytosis** (MCV >110 fL).
- **Lactate dehydrogenase (LDH)** and **unconjugated bilirubin** raised due to ineffective erythropoiesis.
- **Bone marrow**: hypercellular with **megaloblastic changes** (abnormally large RBC precursors with immature nuclei).
- **Serum levels of methylmalonic acid and homocysteine** are elevated in **cobalamin deficiency**, while only **homocysteine** is elevated in **folate deficiency**.
- **Schilling test** detects **malabsorption** of cobalamin. **Autoantibodies** against intrinsic factor and parietal cells are found in **pernicious anaemia**.

Management

- **Supportive therapy** and **specific therapy**.

a. Supportive therapy

- **Blood transfusion** for severely anaemic and symptomatic patients.
- **Precautions** against circulatory overload, especially in elderly. **IV furosemide** 20-40 mg may be given before transfusion.
- **Treatment of infections** and **cardiac failure**.

b. Specific therapy

- Treatment of underlying cause of **vitamin B12** or **folate deficiency**.
- **Cobalamin deficiency: Parenteral therapy** with **intramuscular cyanocobalamin**. Treatment starts with **1000 µg per week** for 8 weeks, followed by **1000 µg per month**. For **pernicious anaemia**, treatment is lifelong.
- **Folate deficiency**: Oral dose of **1 mg per day**, or up to **5 mg per day** in cases of **malabsorption**.

Q1. Describe aetiology, clinical features, diagnosis and treatment of scurvy or vitamin c deficiency.

Scurvy

- **Systemic disease** caused by **vitamin C deficiency**.
- Leads to **defective vascular integrity** and **wound repair**.
- Common in **early childhood** and **geriatric patients**.
- **Sources:** Citrus fruits, tomatoes.

Aetiology

- Seen in **third world countries** due to poor nutrition.

Clinical Features

- **Anaemia:**
 - **Normocytic, normochromic** type.
- **Delayed wound healing:**
 - Due to poor **collagen synthesis** and infection localization.
- **Haemorrhagic diathesis:**
 - **Bleeding tendency**, haemarthrosis, GI bleeding.
 - Haemorrhages in **skin, mucosa, gums, muscles, joints, periosteum**.
- **Teeth and gum lesions:**
 - Soft, swollen, bleeding gums (**scorbutic gingivitis**).
 - Affects **dentin development**.
- **Skeletal lesions:**
 - Deranged **osteoid matrix formation**.
 - Painful limbs, **costochondral junction enlargement**.
- **Folliculosis:**
 - Abnormal lymph follicle count.
- **Scorbutic child:**
 - **Frog-like position** due to **subperiosteal haemorrhage**.

Oral Manifestations

- Affects **gingiva** and **periodontium**.
- **Atrophy of odontoblasts**, irregular dentin.
- **Swollen, spongy gums (scurvy buds)**.
- Bright red, swollen **interdental/marginal gingiva**.
- **Severe cases:** Periodontal bleeding, **bone loss**, loose teeth.

Management

- **Vitamin C** 250 mg TID orally.

- Treat anaemia with **iron supplements**.
- Adequate **dietary vitamin C** intake reverses scurvy.
- **Vascular changes** are reversible.
- **Periodontal breakdown** is **permanent** (lost bone does not regenerate).

Q11. Discuss aetiology and clinical features of subacute bacterial endocarditis. Add a note on principles of management of Infective Endocarditis prophylaxis.

Microbial infection of heart endothelium, commonly affecting heart valves or endocardium of chambers.

- In arterial sites like **AV shunts, PDA, or coarctation of aorta**, termed **infective endarteritis**.

Pathology

- Damaged **endothelium** (e.g., high-pressure jet injury) leads to platelet-fibrin deposition (**non-bacterial thrombotic endocarditis**).
- Microbes enter bloodstream (via mucosa, skin, infections), colonize deposits, forming **vegetations**.
- **Common organisms**:
 - **Subacute**: *Viridans streptococci* (from oral flora).
 - **Acute**: *Staphylococcus aureus*.
 - **Prosthetic valves**: *Staph. epidermidis*, **HACEK group**, fungi, anaerobes.

Clinical Features

- **General**: Fever, weight loss, night sweats, weakness.
- **Cardiac**: New murmur, heart failure, heart blocks.
- **Extracardiac**:
 - **Signs**: Anaemia, clubbing, splenomegaly, petechiae, **Janeway lesions, Osler nodes, Roth spots**.
 - **Neurological**: Meningitis, embolic infarcts, mycotic aneurysm rupture.
 - **Other**: Glomerulonephritis, septic emboli, organ infarcts.

Investigations

- **Blood culture**: Identifies organism, guides therapy.
- **Echocardiography**: Detects vegetations, abscess, heart function.
 - **Transoesophageal echocardiography**: More sensitive.
- Other findings: Anaemia, leukocytosis, high ESR/CRP, haematuria.

Duke's Criteria for Diagnosis

- **Major**: Positive blood culture, echocardiographic findings, new regurgitation.

- **Minor:** Fever, embolic/vasculitic events, predisposing conditions, suggestive findings.
- **Definite IE:** 2 major or 1 major + 3 minor.
- **Possible IE:** 1 major + 1 minor or 3 minor.

Treatment Principles

- **Antibiotics:**
 - Must be **bactericidal**, parenteral, prolonged duration.
 - Based on culture sensitivity and **MIC values**.
- **Empirical Therapy:**
 - Subacute: **Ceftriaxone + gentamicin**.
 - Prosthetic valves: Add **vancomycin**.
- Common antibiotics:
 - Penicillin G, Gentamicin, Ceftriaxone, Ampicillin, Vancomycin.

Surgical Indications

- Needed for **prosthetic valve**, fungal endocarditis, **heart failure**, abscess, large vegetations.

Prophylaxis

- High-risk patients (valvular/congenital heart diseases) before **bacteraemia-inducing procedures**.
- **Antibiotic Regimen:**
 - Standard: **Amoxicillin 2g oral**, 1 hr before.
 - Penicillin allergy: **Clarithromycin 500mg** or **Clindamycin 600mg oral**.
 - Unable to take oral meds: **Ampicillin IV** or **Clindamycin IV**.

Q4. Describe the etiology, clinical features and management of Diphtheria.

Cause

- Caused by **Corynebacterium diphtheriae**, a gram-positive rod with **Chinese letter configuration**.
- Produces a **powerful exotoxin** damaging the **heart muscle** and **nervous system**.

AetiopathogenesisDiphtheria

- Humans are the only **reservoir**.
- **Transmission:** Aerosol inhalation and direct contact.
- **Diphtheria toxin** causes **tissue necrosis**.

Clinical Features

- **Incubation period:** 2–4 days (average 1 week).
- **Forms:** **Nasal, pharyngeal (most common), laryngeal, cutaneous.**
- **Pharyngeal form:**
 - Low-grade fever, **pharyngeal erythema**, and **pseudomembrane** formation.
- **Laryngeal involvement:** Hoarseness, stridor, **respiratory obstruction.**
- **Nasal diphtheria:** Nasal discharge.
- Other symptoms: **Otitis media, conjunctivitis, keratitis, myocarditis, neuropathies.**
- **Severe cases:** May cause **acute circulatory failure** and death.

Diagnosis

- **Gram's stain**
- **Bacterial culture**
- Clinical identification of **pseudomembrane** and isolation of *C. diphtheriae*.

Treatment

1. **Isolation of patient.**
2. **Antitoxin administration:** Neutralizes free toxin.
 - **Mild cases:** 4,000–8,000 units IM.
 - **Moderate cases:** 16,000–40,000 units IM.
 - **Severe cases:** Up to 100,000 units IV.
3. **Antibiotics:** Prevents spread of *C. diphtheriae*.
 - **Amoxicillin:** 500 mg 8th hourly for 10–14 days.
 - **Erythromycin:** 500 mg 6th hourly.
 - **Clarithromycin, Azithromycin:** Equally effective.
4. **Supportive measures:**
 - **Tracheostomy** in respiratory obstruction.
 - **Immunization** with diphtheria toxoid after recovery.

Prevention

- **Contact tracing, surveillance, prophylactic antibiotics.**
- Routine **immunization.**

Complications

1. **Cardiac:**
 - **Toxic myocarditis:** Tachycardia, feeble heart sounds, cardiac enlargement, tic-tac rhythm, arrhythmias.
 - Sudden death from **congestive heart failure.**
2. **Respiratory:**

- Bronchitis, bronchopneumonia, **respiratory obstruction, respiratory paralysis.**
- 3. **Neurological:**
 - **Paralysis** of palate, accommodation, facial muscles, bulbar muscles, respiratory muscles.
- 4. **Renal:**
 - **Toxic nephritis.**
- 5. **Vascular:**
 - **Middle cerebral artery thrombosis**, monoplegia.
- 6. Other: **Otitis media, arthritis.**